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1 Basic Concepts of Lie Algebra

1.1 Definitions and first examples

Definition 1.1.1

A vector space L over a field $\mathbb{F}$, with an operation $L \times L \to L$, denoted $(x, y) \mapsto [xy]$ and called the ***bracket*** or ***commutator*** of x and y, is called a ***Lie algebra*** over $\mathbb{F}$ if the following axioms are satisfied:
(L1) The bracket operation is bilinear.
(L2) $[xx] = 0$ for all $x \in L$.
(L3) $[x[yz]] + [y[zx]] + [z[xy]] = 0 \quad (x, y, z \in L)$.

Remark.
- (L1) and (L2), applied to $[x + y, x + y]$ imply anticommutativity : (L2′) $[xy] = [-yx]$.
- Conversely, if $\operatorname{char} \mathbb{F} \neq 2$, it is clear that (L2′) will imply (L2).

1.2 Lie algebras of derivations

Definition 1.2.1

- By an ***F-algebra*** (not necessarily associative) we simply mean a vector space $\mathfrak{A}$ over F endowed with a bilinear operation $\mathfrak{A} \times \mathfrak{A} \to \mathfrak{A}$, usually denoted by juxtaposition (unless $\mathfrak{A}$ is a Lie algebra, in which case we always use the bracket).
- By a ***derivation*** of $\mathfrak{A}$ we mean a linear map $\delta : \mathfrak{A} \to \mathfrak{A}$ satisfying the familiar product rule $\delta(ab) = a\delta(b) + \delta(a)b$ for each $n \in \mathbb{Z}$.